

Supplementary Information

Quantum annealing penalized fusion expectation-maximization for Gaussian mixture estimation

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1 Simulation-generating models and unfavorable initializations

All main simulations used $n = 500$ observations and 30 Monte Carlo replicates. Within a model, every method received exactly the same initialization. The initial means were intentionally concentrated relative to the generating means and the initial variances or covariance matrices were inflated. Tables S1–S3 give the complete settings. Here “T” denotes a generating (true) component and “I” an initialized component. Covariance matrices are symmetric.

Table 1: One-dimensional generating parameters and initial values.

Model	Set	Component	π_k	μ_k	σ_k^2
1	T	1	0.42	-1.50	1.25
1	T	2	0.58	1.50	0.85
1	I	1	0.50	-0.20	4.00
1	I	2	0.50	0.20	4.00
2	T	1	0.20	-4.40	1.20
2	T	2	0.30	-1.40	0.75
2	T	3	0.30	1.40	0.75
2	T	4	0.20	4.40	1.20
2	I	1	0.25	-0.90	5.00
2	I	2	0.25	-0.30	5.00
2	I	3	0.25	0.30	5.00
2	I	4	0.25	0.90	5.00
3	T	1	0.14	-6.00	1.20
3	T	2	0.17	-3.60	0.90
3	T	3	0.19	-1.20	0.75
3	T	4	0.19	1.20	0.75
3	T	5	0.17	3.60	0.90
3	T	6	0.14	6.00	1.20
3	I	1	1/6	-1.25	6.00
3	I	2	1/6	-0.75	6.00
3	I	3	1/6	-0.25	6.00
3	I	4	1/6	0.25	6.00
3	I	5	1/6	0.75	6.00
3	I	6	1/6	1.25	6.00

Table 2: Two-dimensional generating parameters and initial values.

Model	Set	Component	π_k	$\mu_k^\top$	Σ_k
1	T	1	0.40	$(-2.0, 1.0)$	$\begin{bmatrix} 2.2 & 1.4 \\ 1.4 & 1.5 \end{bmatrix}$
1	T	2	0.60	$(1.6, -0.6)$	$\begin{bmatrix} 1.2 & -0.7 \\ -0.7 & 1.4 \end{bmatrix}$
1	I	1	0.50	$(0, -0.25)$	$7.5I_2$
1	I	2	0.50	$(0, 0.25)$	$7.5I_2$
2	T	1	0.20	$(-4.2, 1.5)$	$\begin{bmatrix} 1.2 & 0.35 \\ 0.35 & 1.0 \end{bmatrix}$
2	T	2	0.30	$(-1.4, -0.8)$	$\begin{bmatrix} 0.9 & -0.25 \\ -0.25 & 0.8 \end{bmatrix}$
2	T	3	0.30	$(1.4, 0.8)$	$\begin{bmatrix} 0.9 & 0.25 \\ 0.25 & 0.8 \end{bmatrix}$
2	T	4	0.20	$(4.2, -1.5)$	$\begin{bmatrix} 1.2 & -0.35 \\ -0.35 & 1.0 \end{bmatrix}$
2	I	1	0.25	$(-0.9, 0.2)$	$5I_2$
2	I	2	0.25	$(-0.3, -0.1)$	$5I_2$
2	I	3	0.25	$(0.3, 0.1)$	$5I_2$
2	I	4	0.25	$(0.9, -0.2)$	$5I_2$
3	T	1	0.14	$(-5.6, 1.8)$	$\begin{bmatrix} 1.15 & 0.35 \\ 0.35 & 1.05 \end{bmatrix}$
3	T	2	0.17	$(-3.3, -1.6)$	$\begin{bmatrix} 0.9 & -0.3 \\ -0.3 & 0.9 \end{bmatrix}$
3	T	3	0.19	$(-1.1, 1.2)$	$\begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix}$
3	T	4	0.19	$(1.1, -1.2)$	$\begin{bmatrix} 0.8 & -0.2 \\ -0.2 & 0.8 \end{bmatrix}$
3	T	5	0.17	$(3.3, 1.6)$	$\begin{bmatrix} 0.9 & 0.3 \\ 0.3 & 0.9 \end{bmatrix}$
3	T	6	0.14	$(5.6, -1.8)$	$\begin{bmatrix} 1.15 & -0.35 \\ -0.35 & 1.05 \end{bmatrix}$
3	I	1	1/6	$(-1.20, 0.20)$	$6.5I_2$
3	I	2	1/6	$(-0.72, -0.12)$	$6.5I_2$
3	I	3	1/6	$(-0.24, 0.04)$	$6.5I_2$
3	I	4	1/6	$(0.24, -0.04)$	$6.5I_2$
3	I	5	1/6	$(0.72, 0.12)$	$6.5I_2$
3	I	6	1/6	$(1.20, -0.20)$	$6.5I_2$

Table 3: Three-dimensional generating parameters and initial values.

Model	Set	Component	π_k	$\mu_k^\top$	Σ_k
1	T	1	0.44	$(-1.8, -0.7, 1.0)$	$\begin{bmatrix} 1.9 & 0.9 & -0.55 \\ 0.9 & 1.3 & -0.3 \\ -0.55 & -0.3 & 1.35 \end{bmatrix}$
1	T	2	0.56	$(1.7, 0.6, -0.9)$	$\begin{bmatrix} 1.45 & -0.75 & 0.45 \\ -0.75 & 1.35 & -0.25 \\ 0.45 & -0.25 & 1.2 \end{bmatrix}$
1	I	1	0.50	$(-0.1, 0, 0)$	$8I_3$
1	I	2	0.50	$(0.1, 0, 0)$	$8I_3$
2	T	1	0.20	$(-4.2, 1.5, 0.25)$	$\begin{bmatrix} 1.2 & 0.35 & 0 \\ 0.35 & 1.0 & 0 \\ 0 & 0 & 1.1 \end{bmatrix}$
2	T	2	0.30	$(-1.4, -0.8, -0.25)$	$\begin{bmatrix} 0.9 & -0.25 & 0 \\ -0.25 & 0.8 & 0 \\ 0 & 0 & 1.1 \end{bmatrix}$
2	T	3	0.30	$(1.4, 0.8, 0.25)$	$\begin{bmatrix} 0.9 & 0.25 & 0 \\ 0.25 & 0.8 & 0 \\ 0 & 0 & 1.1 \end{bmatrix}$
2	T	4	0.20	$(4.2, -1.5, -0.25)$	$\begin{bmatrix} 1.2 & -0.35 & 0 \\ -0.35 & 1.0 & 0 \\ 0 & 0 & 1.1 \end{bmatrix}$
2	I	1	0.25	$(-0.9, 0.2, 0)$	$5I_3$
2	I	2	0.25	$(-0.3, -0.1, 0)$	$5I_3$
2	I	3	0.25	$(0.3, 0.1, 0)$	$5I_3$
2	I	4	0.25	$(0.9, -0.2, 0)$	$5I_3$
3	T	1	0.14	$(-5.6, 1.8, 1.4)$	$\text{diag}(1.15, 1.05, 0.9)$
3	T	2	0.17	$(-3.3, -1.6, -1.2)$	$\text{diag}(0.9, 0.9, 0.75)$
3	T	3	0.19	$(-1.1, 1.2, 1.0)$	$\text{diag}(0.8, 0.8, 0.7)$
3	T	4	0.19	$(1.1, -1.2, -1.0)$	$\text{diag}(0.8, 0.8, 0.7)$
3	T	5	0.17	$(3.3, 1.6, 1.2)$	$\text{diag}(0.9, 0.9, 0.75)$
3	T	6	0.14	$(5.6, -1.8, -1.4)$	$\text{diag}(1.15, 1.05, 0.9)$
3	I	1	1/6	$(-1.5, 0.2, 0)$	$7.5I_3$
3	I	2	1/6	$(-0.9, -0.15, 0)$	$7.5I_3$
3	I	3	1/6	$(-0.3, 0.05, 0)$	$7.5I_3$
3	I	4	1/6	$(0.3, -0.05, 0)$	$7.5I_3$
3	I	5	1/6	$(0.9, 0.15, 0)$	$7.5I_3$
3	I	6	1/6	$(1.5, -0.2, 0)$	$7.5I_3$

2 Algorithm settings and reproducibility

QAPF-EM used $\beta_0 = 0.82$, $\rho_\beta = 1.018$, $\Gamma_0 = 0.10$, $\rho_\Gamma = 0.92$, $\gamma = 0.8$, and $a = 3$. The BIC-type grid search selected $\lambda = 4 \times 10^{-4}$ and $\kappa = 0.8$ for the principal analysis. The maximum iteration count was 200. The common baseline tolerance was 10^{-6} and the baseline covariance lower bound was 10^{-5} . DAEM used $\beta_0 = 0.55$ and $\rho_\beta = 1.025$. SEM sampled latent labels on every iteration. SAEM used a 20-iteration warm-up and $r_t = (t - 18)^{-0.6}$ thereafter.

For the simulated two-, four-, and six-component models, data seeds were 13001–13030, 14001–14030, and 15001–15030, respectively. Stochastic-method seeds were 23001–23030, 24001–24030, and 25001–25030. For the real-data experiments, SEM and SAEM used seeds 94001–94030; the maximum iteration count was 200, tolerance 10^{-5} , and covariance loading 10^{-6} . Deterministic methods were repeated as a reproducibility check and returned identical results under fixed data, initialization, and tuning parameters.

The main nine-setting comparison initialized each method with the true number of components and therefore evaluates robustness and parameter recovery, not model-order selection. Only the penalty-sensitivity experiment deliberately over-specified the initial count ($K_{\text{init}} = 7$ for a four-component generating mixture). In that experiment, the active count is the number of positive-weight components after optimization; the displayed effective count also merges estimated means less than 0.16 apart.

3 Complete numerical results

Accuracy and ARI are larger-is-better metrics. NLL and the three mean-squared errors are smaller-is-better metrics. Estimated components were matched to generating components by the permutation minimizing total squared mean distance before parameter errors were computed. Values below are mean $\pm$ standard deviation over 30 replicates.

Table 4: Complete one-dimensional simulation results.

Model	Method	Accuracy	ARI	NLL	MSE(π)	MSE(μ)	MSE(Σ)
1	CEM	.9246 $\pm$.0204	.7218 $\pm$.0667	967.5826 $\pm$ 11.3934	.0026 $\pm$.0038	.0424 $\pm$.0555	.0849 $\pm$.0459
1	DAEM	.9247 $\pm$.0130	.7212 $\pm$.0442	989.4597 $\pm$ 22.7943	.0051 $\pm$.0024	1.6855 $\pm$ 1.0222	3.4815 $\pm$ 2.2284
1	SEM	.9167 $\pm$.0266	.6963 $\pm$.0847	964.0836 $\pm$ 11.5629	.0060 $\pm$.0100	.0603 $\pm$.0711	.0696 $\pm$.0907
1	SAEM	.8813 $\pm$.1075	.6238 $\pm$.2344	969.4982 $\pm$ 19.5683	.0055 $\pm$.0065	.2993 $\pm$.6158	.8040 $\pm$ 1.6669
1	QAPF-EM	.9244 $\pm$.0150	.7205 $\pm$.0506	963.5891 $\pm$ 11.6987	.0021 $\pm$.0027	.0265 $\pm$.0233	.0232 $\pm$.0232
2	CEM	.5263 $\pm$.0214	.4378 $\pm$.0154	1292.2770 $\pm$ 11.1532	.0691 $\pm$.0076	1.8963 $\pm$.2211	1.4654 $\pm$.2313
2	DAEM	.5235 $\pm$.0191	.4347 $\pm$.0114	1281.6676 $\pm$ 14.9638	.0025 $\pm$.0000	10.6807 $\pm$.0347	79.6809 $\pm$ 10.3602
2	SEM	.6944 $\pm$.1417	.4590 $\pm$.1892	1254.7309 $\pm$ 11.9036	.0392 $\pm$.0326	1.3127 $\pm$ 1.6670	9.4594 $\pm$ 10.3446
2	SAEM	.6489 $\pm$.0880	.4678 $\pm$.0998	1269.0961 $\pm$ 11.4250	.0110 $\pm$.0070	1.7085 $\pm$.8152	14.8364 $\pm$ 7.1151
2	QAPF-EM	.7681 $\pm$.1093	.5697 $\pm$.1043	1257.6722 $\pm$ 11.8918	.0075 $\pm$.0042	.7432 $\pm$.5878	7.5457 $\pm$ 5.6429
3	CEM	.4249 $\pm$.0264	.3173 $\pm$.0147	1394.8440 $\pm$ 9.1644	.0393 $\pm$.0036	4.4445 $\pm$.2305	2.2905 $\pm$.3447
3	DAEM	.4135 $\pm$.0269	.3144 $\pm$.0133	1401.0732 $\pm$ 9.5770	.0004 $\pm$.0000	16.8098 $\pm$.0323	224.2707 $\pm$ 17.9238
3	SEM	.5276 $\pm$.0821	.3754 $\pm$.0947	1369.4733 $\pm$ 7.6075	.0346 $\pm$.0164	2.0840 $\pm$ 1.3452	57.0103 $\pm$ 34.2161
3	SAEM	.5899 $\pm$.0613	.4715 $\pm$.0526	1373.1061 $\pm$ 6.9060	.0055 $\pm$.0039	1.7112 $\pm$.6816	15.3150 $\pm$ 6.9281
3	QAPF-EM	.6032 $\pm$.0357	.4789 $\pm$.0463	1370.5432 $\pm$ 6.6048	.0046 $\pm$.0034	2.4549 $\pm$.9919	15.5558 $\pm$ 6.2143

Table 5: Complete two-dimensional simulation results.

Model	Method	Accuracy	ARI	NLL	MSE(π)	MSE(μ)	MSE(Σ)
1	CEM	.9138 $\pm$.1036	.7252 $\pm$.2815	1801.0721 $\pm$ 67.4793	.0033 $\pm$.0070	.1781 $\pm$.3208	.7720 $\pm$ 1.4429
1	DAEM	.9668 $\pm$.0066	.8712 $\pm$.0248	1764.8297 $\pm$ 14.6652	.0007 $\pm$.0007	.0113 $\pm$.0174	.0214 $\pm$.0142
1	SEM	.9663 $\pm$.0072	.8693 $\pm$.0272	1765.4447 $\pm$ 14.5173	.0009 $\pm$.0013	.0116 $\pm$.0094	.0248 $\pm$.0193
1	SAEM	.9675 $\pm$.0070	.8737 $\pm$.0266	1764.8480 $\pm$ 14.6641	.0007 $\pm$.0008	.0112 $\pm$.0167	.0214 $\pm$.0146
1	QAPF-EM	.9669 $\pm$.0068	.8715 $\pm$.0256	1764.8297 $\pm$ 14.6652	.0007 $\pm$.0007	.0113 $\pm$.0173	.0214 $\pm$.0142
2	CEM	.6213 $\pm$.0870	.4576 $\pm$.0910	2028.8843 $\pm$ 23.2402	.0396 $\pm$.0134	.8107 $\pm$.3217	.8355 $\pm$.2896
2	DAEM	.7277 $\pm$.1885	.5476 $\pm$.2663	2040.9145 $\pm$ 83.0160	.0028 $\pm$.0038	2.7851 $\pm$ 2.7373	9.8647 $\pm$ 9.9791
2	SEM	.8885 $\pm$.0999	.7698 $\pm$.1402	1972.5660 $\pm$ 26.8799	.0088 $\pm$.0167	.2304 $\pm$.5092	.7685 $\pm$ 2.0611
2	SAEM	.7483 $\pm$.1240	.5732 $\pm$.1484	1997.0062 $\pm$ 31.0747	.0120 $\pm$.0087	.6670 $\pm$.5819	2.4635 $\pm$ 2.6438
2	QAPF-EM	.9193 $\pm$.0970	.8203 $\pm$.1139	1966.0938 $\pm$ 25.3410	.0020 $\pm$.0050	.1264 $\pm$.5095	.1838 $\pm$.6194
3	CEM	.4483 $\pm$.0692	.3271 $\pm$.0686	2306.0557 $\pm$ 33.3604	.0422 $\pm$.0083	2.0711 $\pm$.3516	1.9118 $\pm$ 2.3258
3	DAEM	.7325 $\pm$.1546	.6210 $\pm$.1842	2205.1602 $\pm$ 64.1976	.0037 $\pm$.0034	2.0878 $\pm$ 2.9615	8.9606 $\pm$ 14.3835
3	SEM	.8464 $\pm$.0942	.7468 $\pm$.1211	2170.2509 $\pm$ 29.0482	.0069 $\pm$.0061	.7022 $\pm$ 1.0696	2.3549 $\pm$ 3.2914
3	SAEM	.7143 $\pm$.0843	.5793 $\pm$.0969	2204.2501 $\pm$ 31.8173	.0040 $\pm$.0027	1.2003 $\pm$.8321	4.7585 $\pm$ 4.4915
3	QAPF-EM	.8411 $\pm$.1041	.7456 $\pm$.1234	2167.9558 $\pm$ 29.9390	.0035 $\pm$.0038	.6118 $\pm$.6708	2.3973 $\pm$ 5.9176

Table 6: Complete three-dimensional simulation results.

Model	Method	Accuracy		ARI		NLL		MSE(π)		MSE(μ)		MSE(Σ)	
1	CEM	.9589 $\pm$.0094	.8421 $\pm$.0346	2536.2907 $\pm$ 22.5282	.0017 $\pm$.0015	.0132 $\pm$.0092	.0333 $\pm$.0136						
1	DAEM	.9584 $\pm$.0101	.8405 $\pm$.0370	2533.8881 $\pm$ 22.4681	.0008 $\pm$.0011	.0093 $\pm$.0083	.0208 $\pm$.0194						
1	SEM	.9569 $\pm$.0114	.8350 $\pm$.0410	2534.7954 $\pm$ 22.4842	.0010 $\pm$.0011	.0136 $\pm$.0142	.0303 $\pm$.0405						
1	SAEM	.9576 $\pm$.0106	.8376 $\pm$.0387	2533.9227 $\pm$ 22.4784	.0008 $\pm$.0013	.0102 $\pm$.0098	.0219 $\pm$.0213						
1	QAPF-EM	.9583 $\pm$.0101	.8400 $\pm$.0372	2533.8881 $\pm$ 22.4681	.0008 $\pm$.0011	.0093 $\pm$.0083	.0208 $\pm$.0193						
2	CEM	.6069 $\pm$.0824	.4409 $\pm$.0812	2759.4695 $\pm$ 33.6529	.0404 $\pm$.0128	.5740 $\pm$.1609	.4126 $\pm$.1421						
2	DAEM	.8095 $\pm$.1631	.6660 $\pm$.2134	2713.7238 $\pm$ 55.4864	.0070 $\pm$.0100	.5721 $\pm$ 1.0693	1.4839 $\pm$ 3.0621						
2	SEM	.8482 $\pm$.1179	.7164 $\pm$.1505	2699.8773 $\pm$ 33.9061	.0118 $\pm$.0178	.4301 $\pm$.8441	.9652 $\pm$ 4.4034						
2	SAEM	.7337 $\pm$.1200	.5482 $\pm$.1386	2726.5540 $\pm$ 36.1804	.0097 $\pm$.0083	.5488 $\pm$.4299	1.1244 $\pm$ 1.2537						
2	QAPF-EM	.8564 $\pm$.1202	.7240 $\pm$.1588	2699.7840 $\pm$ 31.6540	.0082 $\pm$.0117	.2673 $\pm$.4356	.8602 $\pm$ 1.5641						
3	CEM	.5664 $\pm$.0918	.4641 $\pm$.1339	2976.4301 $\pm$ 56.8326	.0274 $\pm$.0092	1.4866 $\pm$.3140	1.1597 $\pm$.3077						
3	DAEM	.6677 $\pm$.1308	.5407 $\pm$.1578	2894.5010 $\pm$ 42.6712	.0076 $\pm$.0047	1.2500 $\pm$.8099	5.8654 $\pm$ 4.7611						
3	SEM	.7871 $\pm$.1048	.6851 $\pm$.1306	2863.0523 $\pm$ 33.4873	.0103 $\pm$.0058	1.1091 $\pm$.9029	2.3903 $\pm$ 4.5022						
3	SAEM	.6914 $\pm$.0854	.5675 $\pm$.0997	2904.3829 $\pm$ 38.0044	.0052 $\pm$.0027	1.1185 $\pm$.6414	3.6862 $\pm$ 2.5411						
3	QAPF-EM	.8170 $\pm$.1392	.7309 $\pm$.1729	2859.5328 $\pm$ 38.6070	.0042 $\pm$.0052	.6451 $\pm$.7004	1.3859 $\pm$ 2.0030						

Table 7: Complete real-data results. Deterministic methods have standard deviation zero. Parameter errors use empirical labelled-class parameters as descriptive references.

Data	Method	Accuracy	ARI	NLL	MSE(π)	MSE(μ)	MSE(Σ)
Iris	CEM	.6867	.5621	300.6287	.0655	.0858	.0121
Iris	DAEM	.8200	.6007	299.8482	.0003	.0472	.0264
Iris	SEM	.7684 $\pm$.1014	.6137 $\pm$.1074	300.6705 $\pm$ 14.3860	.0364 $\pm$.0254	.0866 $\pm$.0604	.0546 $\pm$.1376
Iris	SAEM	.7364 $\pm$.0697	.5448 $\pm$.0620	308.6257 $\pm$ 21.4912	.0215 $\pm$.0202	.0787 $\pm$.0481	.0486 $\pm$.0755
Iris	QAPF-EM	.8200	.6007	299.8482	.0003	.0470	.0263
Wine	CEM	.6124	.3658	2169.5467	.0756	.1555	.1793
Wine	DAEM	.7191	.4419	2131.6593	.0244	.1157	.0492
Wine	SEM	.6307 $\pm$.0751	.3403 $\pm$.0707	2149.2729 $\pm$ 41.4477	.0338 $\pm$.0170	.1760 $\pm$.0744	.0855 $\pm$.0306
Wine	SAEM	.6159 $\pm$.0762	.3025 $\pm$.0950	2181.6745 $\pm$ 49.6358	.0247 $\pm$.0151	.1773 $\pm$.0674	.0890 $\pm$.0257
Wine	QAPF-EM	.7472	.4248	2137.6490	.0030	.0454	.0365
Seeds	CEM	.6810	.4504	-189.7223	.0515	.1122	.0249
Seeds	DAEM	.8952	.7234	-320.3696	.0026	.0164	.0040
Seeds	SEM	.8259 $\pm$.1084	.6431 $\pm$.1365	-253.1673 $\pm$ 46.8323	.0161 $\pm$.0209	.0663 $\pm$.0888	.0487 $\pm$.0923
Seeds	SAEM	.8076 $\pm$.1097	.6093 $\pm$.1390	-229.2785 $\pm$ 74.2504	.0155 $\pm$.0172	.0857 $\pm$.1197	.0534 $\pm$.1318
Seeds	QAPF-EM	.8952	.7234	-320.3696	.0026	.0164	.0040

4 Proofs of the fixed-stage optimization statements

4.1 Validity of the quantum responsibilities

Let $H_{i,\Gamma}$ be Hermitian. It has an eigendecomposition $H_{i,\Gamma} = U \text{diag}(\xi_r) U^\dagger$ with real eigenvalues. Therefore

$$P_i = \exp(-\beta H_{i,\Gamma}) = U \text{diag}\{\exp(-\beta \xi_r)\} U^\dagger \succ 0$$

for $\beta > 0$. Its trace is positive, so $Q_i = P_i / \text{Tr}(P_i)$ is positive semidefinite and has unit trace. For each latent basis vector $|k\rangle$, $\omega_{ik} = \langle k | Q_i | k \rangle \geq 0$. Completeness of the basis gives

$$\sum_{k=1}^K \omega_{ik} = \sum_k \langle k | Q_i | k \rangle = \text{Tr}(Q_i) = 1.$$

Thus the E-step produces a valid probability vector for every observation.

4.2 Conditional monotonicity and objective-value convergence

Fix β , Γ , the active set, the ordering, and both adjacency structures. Let $\tilde{\mathcal{F}}(G; Q)$ denote the penalized variational lower bound and $Q^{(t+1)}$ the exact E-step at $G^{(t)}$. Exactness makes the bound tight:

$$\tilde{\mathcal{F}}(G^{(t)}; Q^{(t+1)}) = \tilde{\mathcal{G}}_{\beta,\Gamma}(G^{(t)}).$$

The accepted generalized M-step satisfies

$$\tilde{\mathcal{F}}(G^{(t+1)}; Q^{(t+1)}) \geq \tilde{\mathcal{F}}(G^{(t)}; Q^{(t+1)}).$$

Because the variational expression is a lower bound for every candidate parameter,

$$\tilde{\mathcal{G}}_{\beta,\Gamma}(G^{(t+1)}) \geq \tilde{\mathcal{F}}(G^{(t+1)}; Q^{(t+1)}) \geq \tilde{\mathcal{F}}(G^{(t)}; Q^{(t+1)}) = \tilde{\mathcal{G}}_{\beta,\Gamma}(G^{(t)}).$$

This proves conditional monotonicity. If the objective is continuous and bounded above on the fixed-structure feasible space, the monotone sequence of objective values converges to a finite limit.

4.3 Conditional stationarity of accumulation points

Let $\mathcal{M}$ be the fixed-stage update map and let $\mathcal{S}$ be the set of directional stationary points. Compactness gives a convergent subsequence $G^{(t_j)} \rightarrow G^*$. Closedness of $\mathcal{M}$ permits a corresponding subsequence $G^{(t_j+1)} \rightarrow \bar{G}$ with $\bar{G} \in \mathcal{M}(G^*)$. Convergence of the objective values and continuity imply

$$\tilde{\mathcal{G}}_{\beta,\Gamma}(\bar{G}) = \tilde{\mathcal{G}}_{\beta,\Gamma}(G^*).$$

If $G^* \notin \mathcal{S}$, the assumed strict-improvement property and first-order consistency of the local approximation would instead give a strict inequality, a contradiction. Hence every accumulation point belongs to $\mathcal{S}$. At an interior differentiable point this is $\nabla_G \tilde{\mathcal{G}}_{\beta,\Gamma}(G^*) = 0$; otherwise it is expressed as

$$0 \in -\partial \tilde{\mathcal{G}}_{\beta,\Gamma}(G^*) + N_{\Theta}(G^*),$$

where ∂ is a generalized subdifferential and N_{Θ} is the normal cone. None of these arguments covers an update that changes the annealing parameters, deletes a component, or changes an adjacency relation.

5 Interpretive limits

The complete results support the claims stated in the main paper but also delimit them. QAPF-EM is not best for every individual metric. In the simplest two-component models, several annealing methods are nearly indistinguishable. The clearest improvements occur in the more difficult four- and six-component settings. Furthermore, the main simulation comparison fixes the candidate count at the true count, while the seven-to-four experiment is a single controlled demonstration of structural simplification. A broader model-order study should vary dimension, separation, sample size, covariance structure, and degree of over-specification and should compare against dedicated order-selection procedures.